chap_grappe_comparaison

2025-07-15

```r
library(sampling)
library(survey)
```

```
## Loading required package: grid

## Loading required package: Matrix

## Loading required package: survival

##
## Attaching package: 'survival'

## The following objects are masked from 'package:sampling':
##
##     cluster, strata

##
## Attaching package: 'survey'

## The following object is masked from 'package:graphics':
##
##     dotchart
```

```r
data(rec99)
```

Comparaison des stratégies basées sur un plan en grappes par étude de Monte-Carlo

Plan en grappes selon un tirage SI (SIC)

```r
nI <- 4; NI <- length(table(rec99$BVQ_N))
nsim <- 10000
est.grapSI <- numeric(nsim)
n.grapSI <- numeric(nsim)
set.seed(123456)
for (i in 1:nsim)
 {
 grap.rec <- sampling::cluster(rec99,clustername=c("BVQ_N"),
 size=nI,method="srswor")
 grap2.rec <- getdata(rec99, grap.rec)
 n.grapSI[i] <- nrow(grap2.rec)
 ech.grap <- svydesign(ids=~BVQ_N, probs=~Prob,
 data=grap2.rec,
 fpc=~rep(nI/NI, n.grapSI[i]))
 est.grapSI[i] <- svytotal(~LOGVAC, ech.grap)[1]
 }
```

```r
mean(est.grapSI)
```

```
## [1] 10791.75
```

```r
#[1] 10791.52
(cv.grapSI <- sd(est.grapSI)/mean(est.grapSI))
```

```
## [1] 0.3159242
```

```r
#[1] 0.3162824
```

```r
quantile(n.grapSI,prob=c(0.05,0.25,0.5,0.75,0.95))
```

```
##  5% 25% 50% 75% 95%
##  40  53  65  81 114
```

```r
#5% 25% 50% 75% 95%
#40 53 65 81 113
range(n.grapSI)
```

```
## [1]  25 161
```

```r
#[1] 24 162
```

Plan en grappes selon un tirage STSI

```r
o <- as.data.frame(table(rec99$BVQ_N))
names(o) <- c("BVQ_N","Ni")
rec_st <- merge(rec99,o)
rec_st$stratBVQ <- ifelse(rec_st$Ni < 20, 1, 2)
NBVQ <- table(rec_st$stratBVQ,rec_st$BVQ_N)
NBVQ <- ifelse(NBVQ==0,0,1)
NBVQ<-apply(NBVQ,1,sum)
n.grappe <- 2
```

```r
n.sim<-10000
est.grapSTSI <- numeric(n.sim)
set.seed(123456)
for (i in 1:n.sim)
 { grap.st.rec <- mstage(rec_st,stage=c("stratified","cluster"),
   varnames=list("stratBVQ","BVQ_N"),
   size=list(table(rec_st$stratBVQ),
   c(n.grappe,n.grappe)),
   method=c("","srswor"))
   grapSTSI.rec <- getdata(rec_st,grap.st.rec)[[2]]
   n_BVQ <- table(grapSTSI.rec$stratBVQ)
   ech.grapSTSI <- svydesign(ids=~BVQ_N,strata =~stratBVQ,
   probs=~Prob,data=grapSTSI.rec,
   fpc=~c(rep(n.grappe/NBVQ[1],n_BVQ[1]),
   rep(n.grappe/NBVQ[2],n_BVQ[2])))
   est.grapSTSI[i] <- svytotal(~LOGVAC, ech.grapSTSI)[1]}
```

```r
mean(est.grapSTSI)
```

```
## [1] 10766.41
```

```r
#[1] 10761.91
(cv.grapSTSI <- sd(est.grapSTSI)/mean(est.grapSTSI))
```

```
## [1] 0.2509657
```

```r
#[1] 0.2544259
```

Plan en grappes proportionnel à la taille

```r
data(swissmunicipalities)
swiss1 <- swissmunicipalities[,c(1,5,6)]
grap.pop <- aggregate(swiss1[,2:3],by=list(swiss1$CT),sum)
nI <- 6
pik.swiss <- inclusionprobabilities(grap.pop$HApoly,nI)
poids <- 1/pik.swiss
```

```r
n.sim <- 10000
est.grap_sy <- numeric(n.sim)
set.seed(123456)
for (i in 1:n.sim)
 {
 sy_swiss <- UPsystematic(pik.swiss)
 ech.sy_grap <- svydesign(id=~1,weights=poids[sy_swiss==1],
 data=grap.pop[sy_swiss==1,])
 est.grap_sy[i] <- svytotal(~Surfacesbois,ech.sy_grap)[1]
 }
```

```r
mean(est.grap_sy)
```

```
## [1] 1270994
```

```r
#[1] 1270994
sd(est.grap_sy)/mean(est.grap_sy)
```

```
## [1] 0.06953361
```

```r
#[1] 0.06953361
```